

Project 1: Healthcare & Telemedicine Infrastructure

Project Title

Data-Driven Optimization of Remote Telemedicine Emergency Allocation Systems

1. Business Problem

Remote healthcare systems often experience:

- Long patient waiting times
- Doctor allocation bottlenecks
- High appointment cancellation rates
- Delayed emergency responses

The objective is to identify factors contributing to treatment delays and recommend improvements to telemedicine operations.

2. Recommended Dataset

Primary Dataset

Medical Appointment No Shows (Kaggle)

Key Attributes:

Feature	Description
PatientId	Unique Patient
AppointmentID	Appointment Number
Gender	Male/Female
ScheduledDay	Booking Date
AppointmentDay	Consultation Date
Age	Patient Age
Neighbourhood	Location
Scholarship	Financial Support
Hypertension	Health Condition
Diabetes	Health Condition
SMS_received	Reminder Sent
No-show	Attended or Not

3. Project Deliverables

Students must submit:

Technical Deliverables

- Google Colab Notebook
- GitHub Repository
- IEEE Research Paper
- Presentation

Business Deliverables

- Executive Summary
 - Telemedicine Insights
 - Improvement Recommendations
-

4. Data Science Workflow

Step 1: Data Collection

```
import pandas as pd

df = pd.read_csv("telemedicine.csv")
```

Step 2: Data Understanding

```
df.head()

df.info()

df.describe()
```

Expected Output:

- Number of Patients
 - Missing Values
 - Data Types
-

Step 3: Missing Value Analysis

```
df.isnull().sum()
```

Questions:

- Which columns contain null values?
 - Can they be removed?
-

Step 4: Duplicate Detection

```
df.duplicated().sum()
```

5. Outlier Analysis (Mandatory)

IQR Method

```
Q1 = df['Age'].quantile(0.25)
```

```
Q3 = df['Age'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5*IQR
```

```
upper = Q3 + 1.5*IQR
```

Visualization:

```
import seaborn as sns
```

```
sns.boxplot(df['Age'])
```

Expected Finding:

- Extreme age records
 - Incorrect patient entries
-

6. Distribution Analysis

Histogram

```
df['Age'].hist()
```

Questions:

- Which age group dominates?
 - Are elderly patients underrepresented?
-

7. Log Transformation

If waiting time is skewed:

```
import numpy as np

df['Waiting_Log'] = np.log1p(df['Waiting_Time'])
```

Compare:

- Before Transformation
 - After Transformation
-

8. Box-Cox Transformation

```
from scipy.stats import boxcox

df['Waiting_Boxcox'], _ = boxcox(df['Waiting_Time'] + 1)
```

Questions:

- Did skewness reduce?
 - Which transformation worked better?
-

9. Univariate Analysis

Analyze:

Age

```
sns.histplot(df['Age'])
```

Gender

```
sns.countplot(x='Gender', data=df)
```

No-show

```
sns.countplot(x='No-show', data=df)
```

10. Bivariate Analysis

Age vs No-show

```
sns.boxplot(  
    x='No-show',  
    y='Age',  
    data=df  
)
```

Questions:

- Which age group misses appointments most?
-

Gender vs No-show

```
pd.crosstab(  
    df['Gender'],  
    df['No-show']  
)
```

11. Correlation Analysis

```
corr = df.corr(numeric_only=True)  
  
sns.heatmap(  
    corr,  
    annot=True  
)
```

Expected Output:

- Strong positive relationships
 - Negative correlations
-

12. Statistical Validation

Chi-Square Test

Question:

Is Gender related to No-show behavior?

```
from scipy.stats import chi2_contingency

table = pd.crosstab(
    df['Gender'],
    df['No-show']
)

chi2, p, dof, exp = chi2_contingency(table)
```

Interpretation:

$p < 0.05$

Significant relationship exists

$p > 0.05$

No significant relationship

ANOVA

Question:

Does waiting time differ across age groups?

```
from scipy.stats import f_oneway
```

Group:

- Young
- Adult
- Senior

Perform ANOVA.

13. Visualizations Required

Students must produce:

Basic

- Histogram
- Bar Chart
- Pie Chart

Intermediate

- Box Plot
- Violin Plot
- Heatmap

Advanced

- Pairplot
 - Correlation Matrix
 - Dashboard Summary
-

14. Business Insights

Students must answer:

Insight 1

Which patient category experiences highest delays?

Insight 2

Do SMS reminders reduce no-shows?

Insight 3

Which neighborhood requires additional doctors?

Insight 4

What causes appointment bottlenecks?

15. Model Building Extensions

Easy

Telemedicine Dashboard

KPIs:

- Total Appointments
- No-show Rate

- Average Waiting Time
-

Medium

Appointment Delay Prediction

Target:

`Delay > 30 mins`

Algorithms:

- Logistic Regression
 - Decision Tree
-

Advanced

Emergency Allocation Recommendation System

Input:

- Patient Severity
- Waiting Time
- Doctor Availability

Output:

- Priority Level
 - Recommended Doctor Allocation
-

16. GitHub Structure

`Healthcare_Telemedicine_Project`

```
|— data
|— notebooks
|— scripts
|— outputs
|— visualizations
|— presentation
|— ieee_report
|— README.md
```

17. IEEE Paper Structure

1. Abstract
 2. Introduction
 3. Literature Review
 4. Methodology
 5. Data Cleaning
 6. Statistical Analysis
 7. Results
 8. Business Recommendations
 9. Conclusion
 10. References
-

18. Mock Interview Questions

Technical

1. Why did you use IQR?
2. Why log transformation?
3. Why Box-Cox?
4. Why Chi-Square?
5. Why ANOVA?

Business

1. What is the biggest bottleneck?
2. How can hospitals reduce delays?
3. What recommendation would you give management?

Communication

1. Explain your project in 60 seconds.
2. Explain one visualization.
3. Explain one business insight.